

Teoremă Fie $A \subset \mathbb{R}^n$ mărginită. Sunt echivalente afirm:

1) $A \in \mathcal{J}(\mathbb{R}^n)$

$$(\text{Fr}(A) = \bar{A} \cap \overline{C A} = \bar{A} \setminus \overset{\circ}{A})$$

2) $\text{Fr}(A) \in \mathcal{J}(\mathbb{R}^n)$ și $\lambda(\text{Fr}(A)) = 0$.

Dem. 1 $\Rightarrow$ 2) Fie $A \in \mathcal{J}(\mathbb{R}^n)$ și fie $\varepsilon > 0$.

Există $P, Q \in \mathcal{E}(\mathbb{R}^n)$ a.î $P \subset A \subset Q$, $\lambda(Q) - \lambda(P) < \varepsilon$.

$$\overset{\circ}{P}, \bar{Q} \in \mathcal{E}(\mathbb{R}^n), \lambda(\overset{\circ}{P}) = \lambda(P), \lambda(\bar{Q}) = \lambda(Q)$$

$$\overset{\circ}{P} \subset A \Rightarrow \overset{\circ}{P} \subset \overset{\circ}{A} \Rightarrow C \overset{\circ}{P} \supset C \overset{\circ}{A} = \overline{C A} \supset \text{Fr}(A) \Rightarrow \text{Fr}(A) \subset C \overset{\circ}{P}$$

$$A \subset Q \Rightarrow \bar{A} \subset \bar{Q} \Rightarrow \text{Fr}(\bar{A}) \subset \bar{Q}$$

$$\Rightarrow \text{Fr}(A) \subset \bar{Q} \cap C \overset{\circ}{P} = \bar{Q} \setminus \overset{\circ}{P}$$

$$\left. \begin{aligned} & \subset \text{Fr}(A) \subset \bar{Q} - \overset{\circ}{P} \\ & \overset{\circ}{P}, \bar{Q} \in \mathcal{E}(\mathbb{R}^n) \Rightarrow \bar{Q} - \overset{\circ}{P} \in \mathcal{E}(\mathbb{R}^n) \\ & \lambda(\bar{Q} \setminus \overset{\circ}{P}) = \lambda(\bar{Q}) - \lambda(\overset{\circ}{P}) < \varepsilon \end{aligned} \right\} \Rightarrow \lambda^*(\text{Fr}(A)) < \varepsilon$$

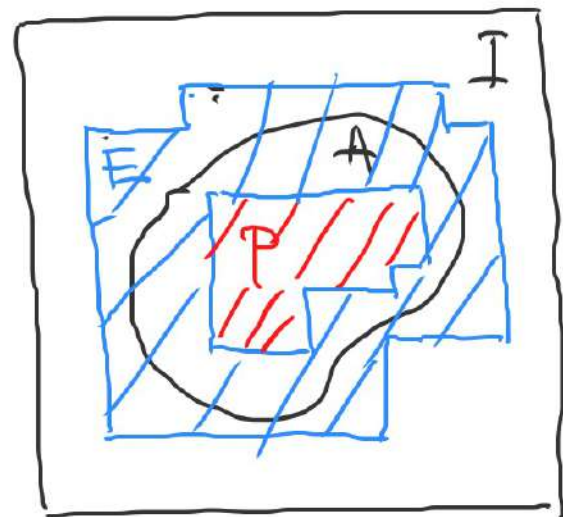
Cum ε a fost ales arbitrar $\Rightarrow \lambda^*(\text{Fr}(A)) = 0$

$$\Rightarrow \lambda_*(\text{Fr}(A)) = \lambda^*(\text{Fr}(A)) = 0 \Rightarrow \text{Fr}(A) \in \mathcal{J}(\mathbb{R}^n) \text{ si } \lambda(\text{Fr}(A)) = 0.$$

2 $\Rightarrow$ 1) $A \subset \mathbb{R}^n$ m\u00e2ngime, $\text{Fr}(A) \in \mathcal{J}(\mathbb{R}^n)$, $\lambda(\text{Fr}(A)) = 0$
 Fie $\varepsilon > 0$. Exist\u00e2 $E \in \mathcal{E}(\mathbb{R}^n)$ a. \u0162

$$\text{Fr}(A) \subset E \quad \text{si} \quad \lambda(E) < \varepsilon.$$

Fie $I \subset \mathbb{R}^n$ interval a. \u0162 $A \cup E \subset I$
 Atunci $I \setminus E \in \mathcal{E}(\mathbb{R}^n)$.



$I \setminus E \in \mathcal{E}(\mathbb{R}^n) \Rightarrow \exists (I_j)_{j \in L}, |L| < \infty$ o familie finita de
intervale din $\mathbb{R}^n$ a.i. $I \setminus E \subset \bigcup_{j \in L} I_j$

$$A \setminus E \subset I \setminus E \Rightarrow A \setminus E = \bigcup_{j \in L} (I_j \cap (A \setminus E))$$

$$K = \{j \in L \mid I_j \cap (A \setminus E) \neq \emptyset\}$$

$$(*) \quad A \setminus E = \bigcup_{j \in K} (I_j \cap (A \setminus E)); \text{ Arăta că } \forall j \in K, I_j \subset A \setminus E.$$

$$\left. \begin{array}{l} \text{Fie } j \in K. \\ I_j \cap E = \emptyset \\ I \supset Fr(A) \end{array} \right\} \Rightarrow I_j \cap Fr(A) = \emptyset \Rightarrow I_j \subset \overset{\circ}{A} \cup \overset{\circ}{\widehat{C}A}.$$

$$\emptyset \neq I_j \cap (A \setminus E) \subset I_j \cap (A \setminus Fr(A)) = I_j \cap \overset{\circ}{A} \Rightarrow I_j \cap \overset{\circ}{A} \neq \emptyset.$$

$$\left. \begin{array}{l} I_j \subset \overset{\circ}{A} \cup \overset{\circ}{C}A \\ I_j \cap \overset{\circ}{A} \neq \emptyset \\ I_j \text{ conexă} \end{array} \right\} \Rightarrow I_j \subset \overset{\circ}{A} \subset A \quad \left. \begin{array}{l} I_j \cap E = \emptyset \Rightarrow I_j \subset C E \\ \end{array} \right\} \Rightarrow I_j \subset A \cap C E = A \setminus E$$

Deci $\forall j \in K$ avem. $I_j \subset A \setminus E \stackrel{(*)}{\Rightarrow} A \setminus E = \bigcup_{j \in K} I_j$

Asadar. $P = A \setminus E \in \mathcal{E}(\mathbb{R}^n)$.

$$Q = P \cup E \in \mathcal{E}(\mathbb{R}^n), \quad P \cap E = (A \setminus E) \cap E = \emptyset$$

$$Q = P \cup E = (A \setminus E) \cup E = A \cup E \supset A.$$

$$P \subset A \subset Q$$

$$\lambda(Q) - \lambda(P) = \lambda(P \cup E) - \lambda(P) = \cancel{\lambda(P)} + \lambda(E) - \cancel{\lambda(P)} < \varepsilon.$$

Asadar exista P, Q elementare si

$$P \subset A \subset Q \text{ si } \lambda(Q) - \lambda(P) < \varepsilon.$$

Cum a fost ales arbitrar rezultă ca $A \in \mathcal{J}(\mathbb{R}^n)$

(rezultă din ultima Propozitie din Cursul 15)

Propozitie Dacă $A \subset \mathbb{R}^n$ este compactă si neglijabilă Lebesgue atunci $A \in \mathcal{J}(\mathbb{R}^n)$ si $\lambda(A) = 0$.

Remarca Fie $A \subset \mathbb{R}^n$ mărginită. Sunt echivalente.

$$(1) A \in \mathcal{J}(\mathbb{R}^n) \text{ si } \lambda(\text{Fr}(A)) = 0 \quad (2) \lambda^*(\text{Fr}(A)) = 0.$$

O altfel de mulțime si numeste neglijabilă Jordan.

Teorema lui Fubini

Fie $A \subset \mathbb{R}^1$, $B \subset \mathbb{R}^2$ intervale, $f: A \times B \rightarrow \mathbb{R}$

$$x \in A : f_x: B \rightarrow \mathbb{R}, f_x(y) = f(x, y)$$

$$y \in B : f_y: A \rightarrow \mathbb{R}, f_y(x) = f(x, y)$$

$$L(x) := \int_B f_x = \int_B f(x, y) dy$$

$$U(x) = \int_B f_x = \int_B f(x, y) dy$$

Teorema (Fubini) Fie $A \subset \mathbb{R}^1$, $B \subset \mathbb{R}^2$ intervale

și $f: A \times B \rightarrow \mathbb{R}$ integrabilă Riemann. Atunci funcțiile L și U sunt integrabile și

$$\iint_{A \times B} f = \int_A L(x) dx = \int_A U(x) dx$$

Adică

$$\iint_{A \times B} f = \int_A \left(\int_B f(x, y) dy \right) dx = \int_A \left(\int_B f(x, y) dy \right) dx,$$

La fel

$$\iint_{A \times B} f = \int_B \left(\int_A f(x,y) dx \right) dy = \int_B \left(\int_A f(x,y) dx \right) dy.$$

Demonstratie Fie Δ_A diviziune a lui A și Δ_B diviziune a lui B .

$\Delta = \Delta_A \times \Delta_B$ diviziune a lui $A \times B$.

Orice diviziune a lui $A \times B$ este de această formă.

$$\Delta_A = \{R_1, R_2, \dots, R_m\}, \Delta_B = \{T_1, T_2, \dots, T_n\}$$

$$m_{R_i \times T_j}(f) = \inf \{ f(x, y) \mid (x, y) \in R_i \times T_j \}$$

$$M_{R_i \times T_j}(f) = \sup \{ f(x, y) \mid (x, y) \in R_i \times T_j \}$$

$$m_{T_j}(f_x) = \inf \{ f(x, y) \mid y \in T_j \}, \quad x\text{-fixat}$$

Arrem. $\forall x \in R_i, m_{R_i \times T_j}(f) \leq m_{T_j}(f_x)$

$$\text{For } x \in R_i : \sum_j m_{R_i \times T_j}(f) \cdot \text{vol}(T_j) \leq \sum_j m_{T_j}(f_x) \text{vol}(T_j) =$$

$$= \Delta_{\Delta_B}(f_x) \leq \int_B f_x = \int_B f(x, y) dy = L(x)$$

$$\forall x \in R_i, \quad \sum_j m_{R_i \times T_j}(f) \cdot \text{vol}(T_j) \leq L(x).$$

$$\Rightarrow \forall i, \quad \sum_j m_{R_i \times T_j}(f) \cdot \text{vol}(T_j) \leq \inf_{x \in R_i} L(x) \cdot \text{vol}(R_i)$$

$$\Rightarrow \forall i, \quad \sum_j m_{R_i \times T_j}(f) \cdot \text{vol}(T_j) \cdot \text{vol}(R_i) \leq \left(\inf_{x \in R_i} L(x) \right) \cdot \text{vol}(R_i)$$

$$\sum_i \sum_j m_{R_i \times T_j}(f) \cdot \text{vol}(R_i \times T_j) \leq \sum_i \left(\inf_{x \in R_i} L(x) \right) \cdot \text{vol}(R_i) = \Delta_{\Delta_A}(f)$$

adica

$$\Delta_{\Delta}(f) \leq \Delta_{\Delta_A}(L). \quad \text{La fel} \quad S_{\Delta}(f) \geq S_{\Delta_A}(U).$$

$$\Delta_{\Delta}(f) \leq \Delta_{\Delta_A}(L) \leq S_{\Delta_A}(L) \leq S_{\Delta_A}(U) \leq S_{\Delta}(f), \quad \forall \Delta.$$

$\uparrow$
 $L \leq U$

$$\sup_{\Delta_A} \Delta_{\Delta_A}(L) = \inf_{\Delta_A} S_{\Delta_A}(L) = \iint_{A \times B} f$$

Deci L este integrabila Riemann si

$$\int_A L(x) dx = \int_A \left(\int_{\overline{B}} f(x, y) dy \right) dx = \iint_{A \times B} f$$

$f: D = [a, b] \times [c, d] \rightarrow \mathbb{R}$ integrabila Riemann

$$\iint_D f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_a^b \left(\int_c^d f(x, y) dy \right) dx.$$

Dacă pt orice x funcția

$y \mapsto f(x, y)$ este integrabilă Riemann

atunci

$$\iint_D f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx.$$

Acest lucru se întâmplă dacă $f: D \rightarrow \mathbb{R}$ este continuă.

Asadar dacă $f: D \rightarrow \mathbb{R}$ continuă atunci

$\forall x \in [a, b]$, $y \rightarrow f(x, y)$ este integrabilă pe $[c, d]$.

$\forall y \in [c, d]$, $x \rightarrow f(x, y)$ este integrabilă pe $[a, b]$

funcțiile

$$x \rightarrow \int_c^d f(x, y) dy = \int_c^d f(x, y) dy$$

$$y \rightarrow \int_a^b f(x, y) dx = \int_a^b f(x, y) dx$$

sunt integrabile și

$$\iint_D f(x,y) dx dy = \int_a^b \left(\int_c^d f(x,y) dy \right) dx = \int_c^d \left(\int_a^b f(x,y) dx \right) dy$$

Calculati $\iint_D (x+y) dx dy$. $D = [0,1] \times [0,2]$.

$f: D \rightarrow \mathbb{R}$ $f(x,y) = x+y$ este continuă

$$\iint_D (x+y) dx dy = \int_0^1 \left(\int_0^2 (x+y) dy \right) dx$$

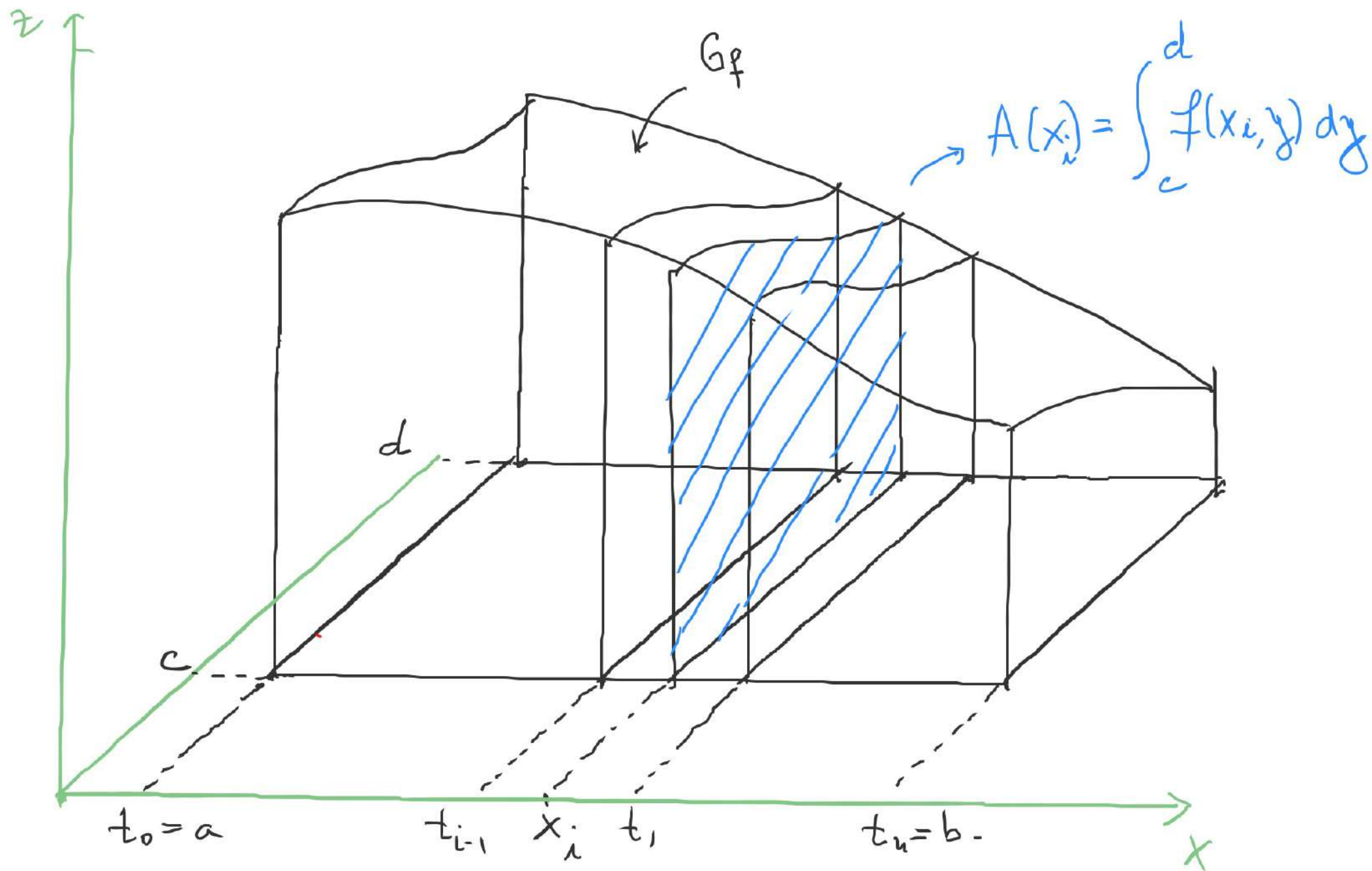
$$\int_0^2 (x+y) dy = \int_0^2 x dy + \int_0^2 y dy = x \int_0^2 dy + \int_0^2 y dy = x \cdot y \Big|_{y=0}^{y=2} + \frac{y^2}{2} \Big|_{y=0}^{y=2}$$

$$\int_0^2 (x+y) dy = 2x+2$$

$$\iint_D (x+y) dx dy = \int_0^1 \left(\int_0^2 (x+y) dy \right) dx = \int_0^1 (2x+2) dx = (x^2+2x) \Big|_0^1 = 3$$

$$\iint_D (x+y) dx dy = \int_0^2 \left(\int_0^1 (x+y) dx \right) dy = \int_0^2 \left(\frac{x^2}{2} + xy \right) \Big|_{x=0}^{x=1} dy$$

$$= \int_0^2 \left(\frac{1}{2} + y \right) dy = \left(\frac{y}{2} + \frac{y^2}{2} \right) \Big|_0^2 = 3.$$



$$\Delta = \{a = t_0 < t_1 < \dots < t_n = b\} \quad t_{i-1} \leq x_i \leq t_i$$

$$\iint_{[a,b] \times [c,d]} f = \sum_{i=1}^n \iint_{[t_{i-1}, t_i] \times [c,d]} f = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n A(x_i) (t_i - t_{i-1})$$

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$$A(x) = \int_c^d f(x, y) dy$$

$x \in [a, b]$

$$\int_a^b A(x) dx$$

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$$\int_a^b \left(\int_c^d f(x, y) dy \right) dx$$